set_block_pin_constraints

NAME

set_block_pin_constraints
Sets pin constraints on the pins of block cells. This command will retire in later releases, please use create_block_pin_constraint instead.

SYNTAX

```
int set_block_pin_constraints
       [-allowed_layers metal_layers]
       [-corner_keepout_num_tracks tracks]
       [-corner_keepout_distance distance]
       [-pin_spacing spacing]
       [-pin_spacing_distance distance]
       [-sides side_numbers]
       [-exclude_sides exclude_sides_numbers]
       [-allow_feedthroughs true | false]
       [-stacking_allowed any | none | with_pg_pins_only | with_signal_pins_only]
       [-cells cell_collection]
       [-width pin_width]
       [-length pin_length]
       [-self]
       [-hard_constraints layer | spacing | location | length | off]
```

Data Types

metal_layers	list
tracks	integer
distance	real
spacing	real
side_numbers	list of integers
exclude_sides_numbers	list of integers
cell_collection	list
pin_width	real or list of layer-real pairs
pin_length	real or list of layer-real pairs

ARGUMENTS

-allowed_layers metal_layers
Specifies the set of metal layers allowed for pin placement. The argument is a list of metal layers.

By default, the metal layers from Min-routing layer to the maximum metal layer (specified earlier during floorplan) are allowed for pin placement. If the maximum metal layer has not been specified, the maximum available metal layer specified in the technology file is used as the default. If the maximum layer specified is beyond the maximum metal layer for the current design, then the maximum metal layer for the current design is used.

-corner_keepout_num_tracks tracks
Specifies the distance in wire tracks from the corners of blocks where pin placement should begin.

If neither the -corner_keepout_num_tracks nor the -corner_keepout_distance option is specified, the default distance is five wire tracks. The value for tracks must be zero or a positive integer. This option is mutually exclusive with option "-corner_keepout_distance". Setting this option invalidates "-corner_keepout_distance" option.

-corner_keepout_distance corner_keepout_distance
Specifies the distance in microns from the corners of blocks where pin placement should begin. The distance has to be 0 or a positive integer. By default, the default distance is five wire tracks. This option is mutually exclusive with the -corner_keepout_num_tracks option. Setting this option invalidates -corner_keepout_num_tracks option.

-pin_spacing int
Specifies the minimum number of wire tracks between adjacent pins or between a pin and a preroute wire that cuts across a block edge in the normal direction. Pins that are created are always snapped to wire tracks. The default pin-to-pin spacing is one wire track. The spacing number must be zero or a positive integer.

-pin_spacing_distance spacing_distance

Specifies the minimum distance between adjacent pins or between a pin and a preroute wire that cuts across a block edge in the normal direction. Pins that are created are always snapped to wire tracks.

-sides side_numbers

Specifies the block sides on which the pin must be placed.

A side number is a positive integer that starts at one, and increments by one as you proceed clockwise around each edge in the shape. Given any rectilinear shape, the leftmost vertical edge is the starting edge (side number 1). If there are multiple leftmost edges, then the edge 1 is the lowest leftmost edge. You cannot specify a value of 0 for this option.

Side number is counted as observed inside the corresponding reference block itself. You can reset the side number to default by specifying the -sides option with an empty string. For example, to reset the side number, enter
-sides {}
This option is mutually exclusive with the -exclude_sides option.

-exclude_sides exclude_side_numbers

Specifies the block edges on which pins cannot be placed. The exclude_side_numbers argument is a list of positive integers. A side number is a positive integer that starts at one, and increments by one as you proceed clockwise around each edge in the shape. Given any rectilinear shape, the leftmost vertical edge is the starting edge (side number 1). If there are multiple leftmost edges, then the edge 1 is the lowest leftmost edge.

For example, to exclude sides 2, 4, and 6, enter
-exclude_sides {2 4 6}

You can reset the excluded side number to default by specifying the -exclude_sides option with an empty string. For example, to reset the excluded sides, enter
-exclude_sides {}

This option is mutually exclusive with -sides option.

-allow_feedthroughs true | false

Specifies whether feedthrough ports can be created in pin placement flow. By default, this option is false and new feedthroughs cannot be created. When feedthroughs are allowed, global routing creates feedthrough routing on blocks as needed. Each top-level net for which a feedthrough port is created is split into a set of new top-level nets and child-level nets, if feedthrough nets are created for the child nets. Directions are assigned for the newly created feedthrough ports. Because new ports are created in the block cell, the netlist is modified as well.

-stacking_allowed any | none | with_pg_pins_only | with_signal_pins_only

Specifies how signal pins can stack with other pins during pin placement. This option takes one of the four allowed values:

- any
  Signal pins can stack with other signal pins on different layers. Signal pins can also stack with power or ground straps, power or ground pins on different layers as well. This is the default.

- none
  Signal pins cannot stack with other signal pins on different layers. Signal pins cannot stack with power or ground traps, power or ground pins on different layers neither.

- with_pg_pins_only
  Signal pins cannot stack with other signal pins on different layers. But signal pins can stack with power or ground straps, power or ground pins on different layers.

- with_signal_pins_only
  Signal pins can stack with other signal pins on different layers. But signal pins cannot stack with power or ground straps, power or ground pins on different layers.

-cells cell_collection

If this option is specified, the constraints are only applied to pins on those blocks in the cell_collection. By default, the constraints are applied to pins on all blocks.

-width pin_width
Specifies the width of the pin. The width is perpendicular to the layer's preferred routing direction. The syntax can be a single real number to indicate the width is applied to all allowed layers (referred to as common pin width). Or alternatively, the width can be specified as a list of layer-real pairs as the following (referred to as per layer pin width):

```
-width {{M2 0.2} {M3 0.2} {M4 0.3} {M5 0.3}}
```

It means that if the pin is to be placed on layer M2 or M3, its width should be 0.2 and if on M4 or M5, 0.3. On other layers, use the default width defined by the tech file.

If you first set the pin width to be common pin width (or per layer pin width), then in a second set_block_pin_constraints set the pin width to be per layer width (or common pin width), then the width specified in the second command overrides the first.

However, if you first set the pin width to be per layer, then in a second command set the pin width again to be per layer but with different layers or different values, then the combined per layer pin width will be the final constraints. For example, the following example:

```
set_block_pin_constraints -cells mid1 -width 0.3
set_block_pin_constraints -cells mid1 -width {{M2 0.3} {M3 0.3}}
```

The first common pin width constraint is overridden by the second per layer pin width. The pin width should be 0.3 on layer M2 or M3, but default width on all other layers.

And for the following example:

```
set_block_pin_constraints -cells mid1 -width {{M3 0.4} {M4 0.4}}
set_block_pin_constraints -cells mid1 -width {{M2 0.3} {M3 0.3}}
```

The pin width should be 0.3 on layer M2 or M3, 0.4 on layer M4, and default width on all other layers.

-length pin_length
Specifies the length of the pin. The length is in parallel to the layer's preferred routing direction. The syntax can be a single real number to indicate the length is applied to all allowed layers (referred to as common pin length). Or alternatively, the length can be specified as a list of layer-real pairs as the following (referred to as per layer pin length):

```
-length {{M2 0.2} {M3 0.2} {M4 0.3} {M5 0.3}}
```

It means that if the pin is to be placed on layer M2 or M3, its length should be 0.2 and if on M4 or M5, 0.3. On unspecified layers, the pin length is derived from the pin width. See the man page of place_pins for details.

-self Applies constraints only to the top-level design. By default, the constraints are applied to block cells inside current top design.

-hard_constraints off | spacing | location | layer | length
Specifies hard constraint types. The supported hard constraint types are spacing, location, layer, or length. If none of the constraints should be hard, you can use the keyword off.

When one particular type of constraint is specified as hard, place_pins honors it without any design rule checks, even if honoring it means one or more design rule violations.

If spacing is specified as hard, then the minimum spacing between applicable pins is at least the value specified by the associated spacing constraints.

If location is specified as hard, then the applicable pin is placed at the exact side and offset specified by the associated pin constraints. If the location constraint is specified as coordinates by set_individual_pin_constraints instead, the coordinates are snapped to the edge unless option -off_edge is presented, then during place_pins the pin is placed at the snapped edge and offset (option -off_edge is not presented) or the specified coordinates (option -off_edge is presented).

If layer is specified as hard, then the applicable pin is placed at the layer specified by the associated layer constraints.

If length is hard, then the applicable pin's length is exactly the value specified by the associated length constraints.

If two constraints conflict and only one of those two constraints is specified as a hard constraint, the other constraint is ignored. For example, you can specify a layer constraint and a side constraint on a pin. When the layer's preferred direction is horizontal and the side is horizontal edge, if the layer constraint set to hard constraint, place_pins places pins on the horizontal layer and vertical edge as there is no legal horizontal wiretrack on horizontal edge.

You can specify more than one hard constraint type.

By default, this option is off.

DESCRIPTION

This command sets pin placement constraints on blocks. The constraints are honored during pin placement and are saved in the design database.

The constraint values that were set by previous set_block_pin_constraints commands are overridden by subsequent calls of this command. However, any new constraints or switches that are set using this command get appended to the existing ones. Example 1: In the following example, the values in the second command overrides the first one, that is, the tool allows pin placement on layers M5 and M7, and not on M4, M6, M8, and M10.

```
prompt> set_block_pin_constraints -self -allowed_layers {M4 M6 M8 M10}
prompt> set_block_pin_constraints -self -allowed_layers {M5 M7}
```

Example 2: In the following example, the constraints set using the second command are added to the ones that are set by the first one, that is, the tool allows pin placement on sides 1 and 3, on layers M5 and M7.

```
prompt> set_block_pin_constraints -self -sides {1 3}
prompt> set_block_pin_constraints -self -allowed_layers {M5 M7}
```

The set_editability command can enable or disable the set_block_pin_constraints command for specified blocks or levels of physical hierarchy. This command has no effect on the blocks that are disabled with the set_editability command.

This command returns 1 on success, 0 otherwise.

EXAMPLES

Following example sets the allowed block pin layers to METAL2 and METAL3, and allows feedthroughs on all the blocks in the design.

```
  prompt> set_block_pin_constraints -allowed_layers [get_layers METAL2 METAL3] \
     -allow_feedthroughs true
```

SEE ALSO

place_pins(2)
create_block_pin_constraint(2)
remove_block_pin_constraints(2)
report_block_pin_constraints(2)
set_individual_pin_constraints(2)
set_editability(2)
